GEOMETRY · ANGLES AND PARALLEL LINES

Parallel Lines & Transversal

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Parallel Lines & Transversal** on Mr. K's Classroom: Geometry → Angles and Parallel Lines → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. If the lines are parallel and ∠1 = 120°, what do you think ∠5 is?

   120°: they are corresponding angles.

2 Explore use the tool and record what you see

All eight angles (transversal angle 55°, lines parallel)

∠1	∠2	∠3	∠4	∠5	∠6	∠7	∠8
125°	55°	55°	125°	125°	55°	55°	125°

Angle pairs

Pick each pair type in the tool.

Pair type	One example	Relationship when m ∥ n
Corresponding	∠2 and ∠6	congruent
Alternate interior	∠3 and ∠6	congruent
Alternate exterior	∠1 and ∠8	congruent
Same-side interior	∠3 and ∠5	supplementary (sum 180°)
Vertical	∠1 and ∠4	congruent
Linear pair	∠1 and ∠2	supplementary

Tilt line n by 10° (transversal angle 55°)

Pair	Measures	Relationship still true?
∠2 and ∠6 (corresponding)	55° and 45°	no
∠3 and ∠5 (same-side interior)	55° and 135°: sum 190°	no
∠1 and ∠4 (vertical)	125° and 125°	yes

3 Write the rule fill in each blank from what you observed

1. When parallel lines are cut by a transversal, corresponding, alternate interior and alternate exterior angles are **congruent**.
2. Same-side interior angles are **supplementary**, so they add to **180°**.
3. Vertical angles are always **congruent**, even when the lines are not parallel.

4. If the lines are not parallel, the pair relationships **no longer hold**.

4 Check no tool: show your work

1. Lines are parallel and $\angle 3 = 72°$. Find $\angle 6$ and $\angle 5$.

   $\angle 6 = 72°$; $\angle 5 = 108°$

2. Corresponding angles measure $(3x + 10)°$ and $(5x - 30)°$. Find x.

   x = 20

3. Same-side interior angles measure $(2x)°$ and $(3x + 20)°$. Find x.

   5x + 20 = 180, so x = 32

4. Alternate interior angles measure 70° and 75°. Are the lines parallel?

   No; they would have to be congruent

5 Reflect

How could you use angles to prove two lines are parallel?

Show that a pair of corresponding (or alternate interior) angles is congruent, or that same-side interior angles are supplementary.